VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
Subject Professor In-charge	Prof. Snehal Andhare
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Experiment Number	14
Experiment Title	Luggage Complaint Portal: Automated Provisioning and Reliability Validation
Objectives (Skill Set / Knowledge Tested / Imparted)	To provision a clean node with Ansible, deploy the application, rerun the automation to show idempotency, run a health check and test rollback and recovery to the previous stable release.

Program	**1. Provisioning design**
	• Clean node: service, unit file, folders and the portal user removed before the run.
	• Provisioning: ansible-playbook installs Java, the user, folders, the release, the systemd service and runs the health check.
	• Idempotency: a second run of the same playbook should report changed=0.
	• Rollback: running the playbook with the previous app_version repoints current.war and restarts the service.
	• Recovery: the unit uses Restart=always, so systemd restarts the service if it is killed.
	2. Commands used
	`AP="ansible-playbook -i ansible/inventory.ini ansible/site.yml -e app_port=8090"`
	`$AP -e app_version=1.0.0 # provision clean node`
	`$AP -e app_version=1.0.0 # idempotency: expect changed=0`
	`$AP -e app_version=1.1.0 # new release`
	`$AP -e app_version=1.0.0 # rollback to previous stable release`
	`sudo systemctl kill -s KILL luggage-portal # recovery test`
	`curl http://localhost:8090/actuator/health`
	3. Evidence
	[Screenshot to be added: Provisioning run on the clean node (PLAY RECAP)]
	[Screenshot to be added: Second run showing changed=0 (idempotency)]
	[Screenshot to be added: Health check output {"status":"UP"}]
	[Screenshot to be added: Rollback: actuator/info before and after, and the releases folder listing]
	[Screenshot to be added: Recovery: service active again after being killed]

Conclusion	The provisioning playbook installs the application as a versioned, health-checked systemd service. Rerunning it changes nothing, and a rollback is done by running it with the previous version.